

A One-Player Exponential Lower Bound for Smoothed Exact Better-Response Dynamics in Unrestricted Congestion Games

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Abstract

We give a short negative resolution of the unrestricted, arbitrary-pivot interpretation of the smoothed exact better-response question for congestion games. For every integer h we construct a one-player acyclic network congestion game with $h + 1$ vertices and $2h$ edges. The player's strategies are the 2^h source-sink paths of a binary layered graph. Under independent continuous perturbations of the edge latencies, all path costs are distinct almost surely; consequently the exact better-response transition graph is the complete acyclic orientation of these 2^h paths by decreasing cost and contains a directed Hamiltonian path of length $2^h - 1$. Taking uniform perturbations gives this lower bound with density parameter $\phi = 1$.

We also record two fixed-start sharpenings. First, with density parameter $\phi = 2$ and a fixed initial path independent of the realized perturbations, an exact better-response trajectory can be forced to take exactly $2^h - 1$ improving moves almost surely. Second, with $\phi = 1$ and a fixed initial path, the same predecessor-in-cost-order pivot rule has an exactly uniform running-time distribution on $\{0, 1, \dots, 2^h - 1\}$, and hence expectation $(2^h - 1)/2$. The construction uses load-independent latencies, so it simultaneously fits the general-latency, degree-zero polynomial-latency, and one-jump step-function perturbation models.

The lower bound is deliberately about arbitrary exact better responses, or equivalently the longest path in the transition graph. It does not lower-bound shortest-path/best-response dynamics: on the same one-player network, exact best response reaches a pure Nash equilibrium in at most one move. Thus the example separates the broad arbitrary-pivot formulation from the narrower best-response formulation.

1 Introduction

Congestion games admit pure Nash equilibria by Rosenthal's potential argument [3]. A standard way to reach such an equilibrium is exact better-response dynamics: repeatedly choose a player who can strictly decrease her cost and move her to an improving strategy. Giannakopoulos, Grosz, and Melissourgos introduced smoothed models for exact local search in congestion games and proved polynomial smoothed convergence for structurally restricted classes, including restrained congestion games and certain compact network congestion games [1]. Their framework measures the expected number of improving local-search steps under independent continuous perturbations of the numerical cost parameters, with the distributions only constrained by a density upper bound ϕ .

The unrestricted question asks whether the same polynomial smoothed behavior persists without the restraint/compactness assumptions. This note shows that the answer is negative for the broad formulation in which the pivoting rule may choose an arbitrary exact improving deviation and the relevant running time is the maximum length of an improving path in the

transition graph. The counterexample is intentionally minimal: one player, an acyclic network, two parallel edges per layer, and load-independent uniform latencies.

There is a useful caveat. In a one-player network congestion game, a best response is just a shortest path. Therefore shortest-path dynamics terminates immediately on our instance. The result should accordingly be read as a lower bound for arbitrary exact better-response dynamics, not for exact best-response dynamics. This distinction matters because the positive network results of [1] are stated for shortest-path dynamics.

Main contribution. For a succinctly represented network with $2h$ edges, the construction has 2^h strategies. With probability one, these strategies can be ordered by decreasing cost and then visited one by one via exact better responses. Thus the smoothed longest-path complexity is $2^h - 1 = 2^{\Omega(m)}$, where $m = 2h$ is the number of edges. This rules out any bound polynomial in the network description size and ϕ for unrestricted arbitrary-pivot exact better-response dynamics.

2 Model and notation

A congestion game has players N , resources R , strategy sets $\Sigma_i \subseteq 2^R$, and latency functions $\kappa_r : \{1, \dots, |N|\} \rightarrow \mathbb{R}_{\geq 0}$. For a strategy profile σ , the load on resource r is

$$\ell_r(\sigma) = |\{i : r \in \sigma_i\}|,$$

and player i 's cost is

$$C_i(\sigma) = \sum_{r \in \sigma_i} \kappa_r(\ell_r(\sigma)).$$

A pure Nash equilibrium is a profile from which no player has a strictly cheaper unilateral deviation.

A network congestion game is the special case in which the resources are directed edges and each player's strategies are source-sink paths. In this note there is only one player. Hence each strategy is a single path P , the only relevant load is 1, and the path cost is simply

$$C(P) = \sum_{e \in P} \kappa_e(1).$$

Every path of smaller cost than the current path is an exact better response.

The smoothed models considered in [1] perturb the numerical latency parameters independently from continuous distributions on $[0, 1]$ whose densities are bounded above by ϕ . We will use only load-one, load-independent costs, so the same construction fits all three latency representations:

- (i) general latencies, by drawing each value $\kappa_e(1)$ independently;
- (ii) polynomial latencies, by using degree-zero polynomials $\kappa_e(x) = \alpha_e$;
- (iii) step-function latencies, by using one jump at load 1 with jump size α_e .

For a realized instance, the exact better-response transition graph has one vertex per strategy profile and a directed edge from a profile to every profile obtainable by one strictly improving unilateral deviation. For one player this means an arc $P \rightarrow Q$ whenever $C(Q) < C(P)$. The arbitrary-pivot or path-independent smoothed running time is the expected length of the longest directed path in this transition graph.

3 The binary layered construction

Fix $h \geq 1$. Let G_h be the directed acyclic multigraph with vertices

$$v_0, v_1, \dots, v_h,$$

and, for each layer $t = 1, \dots, h$, two parallel directed edges from v_{t-1} to v_t , denoted

$$e_{t,0} \quad \text{and} \quad e_{t,1}.$$

There is one player, with source v_0 and sink v_h . Each source-sink path chooses exactly one edge in every layer, and hence can be identified with a bit string

$$x = (x_1, \dots, x_h) \in \{0, 1\}^h,$$

where path P_x uses edge e_{t,x_t} in layer t . Thus

$$|\mathcal{P}_h| = 2^h,$$

where $\mathcal{P}_h$ is the set of source-sink paths. The network description has $h + 1$ vertices and $2h$ edges.

We first need a generic nondegeneracy observation.

Lemma 1 (Distinct path costs). *Assume that the load-one edge latencies $\{\kappa_e(1) : e \in E(G_h)\}$ are mutually independent continuous random variables. Then, with probability one, all 2^h path costs are distinct.*

Proof. Fix two distinct bit strings $x \neq y$. Choose a layer t with $x_t \neq y_t$. Condition on all edge latencies except $\kappa_{e_{t,x_t}}(1)$. The equality $C(P_x) = C(P_y)$ then determines at most one value of the remaining continuous random variable $\kappa_{e_{t,x_t}}(1)$. Hence this equality has conditional probability zero, and therefore unconditional probability zero. A union bound over the finitely many pairs of paths proves the claim. $\square$

Theorem 2 (Almost-sure exponential longest path). *For the one-player network congestion game on G_h , under any independent continuous perturbations of the load-one edge latencies, the exact better-response transition graph has a directed path of length $2^h - 1$ with probability one. Consequently its longest directed path has length exactly $2^h - 1$ with probability one.*

Proof. By Lemma 1, almost surely the path costs are all distinct. On this event, list the 2^h paths in strictly decreasing cost order:

$$C(P^{(1)}) > C(P^{(2)}) > \dots > C(P^{(2^h)}).$$

For every $j < 2^h$, the move $P^{(j)} \rightarrow P^{(j+1)}$ is a legal exact better response, because the new path has strictly smaller cost. Hence

$$P^{(1)} \rightarrow P^{(2)} \rightarrow \dots \rightarrow P^{(2^h)}$$

is a directed path of length $2^h - 1$ in the transition graph.

No directed path can be longer than the number of vertices minus one, because every exact better-response move strictly decreases cost and therefore the transition graph is acyclic. Since the graph has 2^h vertices, the longest directed path has length exactly $2^h - 1$. $\square$

Corollary 3 (Constant-density smoothed lower bound). *There is no polynomial p such that every unrestricted succinct network congestion game has arbitrary-pivot exact better-response smoothed complexity at most $p(\text{input size}, \phi)$.*

More concretely, taking all load-one edge latencies independently uniform on $[0, 1]$ gives $\phi = 1$ and

$$\mathbb{E}[\text{longest exact better-response path length}] = 2^h - 1$$

for a network whose description size is $\Theta(h)$.

Proof. Uniform $[0, 1]$ latencies have density 1, so this is a $\phi = 1$ smoothed instance. The expectation is immediate from Theorem 2, since the longest path length is $2^h - 1$ almost surely. As the network has $O(h)$ vertices and edges, $2^h - 1$ is exponential in the succinct input size. $\square$

Remark 4 (Explicit strategy lists). *The superpolynomial conclusion uses the succinct network representation. If the 2^h paths were listed explicitly as the player's strategy set, then the same lower bound would be only linear in the explicit list size. Thus the example separates succinct unrestricted strategy spaces from explicitly listed strategy spaces.*

4 Fixed starting path variants

The previous theorem is the cleanest formulation for the standard longest-path measure. It starts the hard trajectory at the realized maximum-cost path, which may depend on the realized perturbations. The next result gives a fixed initial path independent of the realized costs, at the price of increasing the density bound from 1 to 2.

Theorem 5 (Fixed start and full exponential length). *For each h , there is a smoothed instance on G_h with density parameter $\phi = 2$ and a fixed initial path P_{0^h} such that an exact better-response trajectory from P_{0^h} has length exactly $2^h - 1$ almost surely.*

Proof. For every layer t , draw the two edge latencies independently as

$$\kappa_{e_{t,0}}(1) \sim \text{Unif}[1/2, 1], \quad \kappa_{e_{t,1}}(1) \sim \text{Unif}[0, 1/2].$$

All densities are bounded by 2. With probability one,

$$\kappa_{e_{t,0}}(1) > \kappa_{e_{t,1}}(1) \quad \text{for every } t.$$

Therefore replacing any 1-edge of a path by the corresponding 0-edge strictly increases cost. In particular, the all-zero path P_{0^h} is the unique maximum-cost path.

On the probability-one event of distinct path costs, order all paths by decreasing cost,

$$P^{(1)} = P_{0^h}, P^{(2)}, \dots, P^{(2^h)}.$$

The pivot rule that moves from the current path to the path of largest cost strictly below the current cost follows this list exactly. Each move is a strict improvement, and the trajectory terminates at the unique minimum-cost path after $2^h - 1$ moves. $\square$

The original uniform construction also gives an exact distributional statement for a fixed start.

Proposition 6 (Fixed start under uniform perturbations). *Suppose all load-one edge latencies are independent uniform $[0, 1]$ random variables, and fix any initial path P_x before the latencies are realized. Under the predecessor-in-cost-order pivot rule, the number T of exact better-response moves from P_x is uniformly distributed on*

$$\{0, 1, \dots, 2^h - 1\}.$$

In particular,

$$\mathbb{E}T = \frac{2^h - 1}{2}, \quad \text{Var}(T) = \frac{2^{2h} - 1}{12},$$

and for every integer $q \in \{0, \dots, 2^h - 1\}$,

$$\mathbb{P}[T \geq q] = \frac{2^h - q}{2^h}.$$

Proof. By Lemma 1, all path costs are distinct almost surely. The predecessor-in-cost-order rule takes exactly $r - 1$ moves when the initial path has rank r among the 2^h paths in increasing cost order.

It remains to show that this rank is uniform. The joint distribution of edge latencies is invariant under independently swapping the labels 0 and 1 in any subset of layers. These swaps act transitively on the 2^h paths. Hence every path has the same rank distribution. Since exactly one path has each rank, the common rank distribution must be uniform on $\{1, \dots, 2^h\}$. Therefore $T = r - 1$ is uniform on $\{0, \dots, 2^h - 1\}$, and the displayed mean, variance, and tail formula are the elementary formulas for a discrete uniform random variable. $\square$

Remark 7 (A pivot-rule dependence). *The lower bounds above are not claims that every pivot rule is slow. In the one-player construction, exact best response moves directly to a minimum-cost path and terminates in at most one step. Also, if a pivot rule chooses uniformly at random among all lower-cost paths, then conditional on starting at rank r its expected remaining time is the harmonic number H_{r-1} ; under the uniform fixed-start model of Proposition 6, the average is $H_{2^h} - 1 = O(h)$. Thus the example is a sharp warning about arbitrary/adversarial exact improving moves, not about all natural pivoting rules.*

5 Compatibility with the latency representations

We spell out why the same construction applies to the three latency models.

Proposition 8 (General, polynomial, and step-function latencies). *The lower bounds in Theorems 2 and 5 and Corollary 3 hold under each of the following smoothed representations:*

- (i) *general latencies;*
- (ii) *polynomial latencies of degree at most d , even with $d = 0$;*
- (iii) *step-function latencies with at most one breakpoint.*

Proof. There is one player, so the only load ever queried is 1. In the general model, set the single relevant value $\kappa_e(1)$ to be the corresponding independent random variable.

In the polynomial model, use the constant polynomial

$$\kappa_e(x) = \alpha_e,$$

where α_e is the corresponding independent random variable. This has degree 0.

In the step-function model, put one jump at load 1 with jump size α_e , so that $\kappa_e(1) = \alpha_e$. The same independent distributions for the jump sizes reproduce exactly the same path-cost process. $\square$

6 Separation from shortest-path dynamics

The construction deliberately exploits the distinction between arbitrary exact better responses and best responses.

Proposition 9 (Best response is trivial on the same instance). *For every realization of the latencies on G_h , exact best-response dynamics from any initial path reaches a pure Nash equilibrium in at most one move.*

Proof. With one player, a pure Nash equilibrium is exactly a minimum-cost source-sink path. A best response from any nonminimum path is a minimum-cost path. After moving to such a path, no strictly improving deviation exists. If the initial path is already minimum-cost, zero moves are needed. $\square$

Thus the exponential lower bound refutes polynomial smoothed bounds for the unrestricted arbitrary-pivot/longest-path formulation, but it does not refute polynomial smoothed bounds for shortest-path dynamics or other efficiently specified pivoting rules. In particular, the example is consistent with positive results for compact network congestion games under shortest-path dynamics.

7 Conclusion

The unrestricted arbitrary-pivot exact better-response problem has exponential smoothed complexity even in a one-player acyclic network congestion game with independent constant-density perturbations. The mechanism is simple: succinct network descriptions can encode exponentially many paths, and arbitrary exact better-response dynamics can traverse them in decreasing cost order. Smoothing removes ties but does not prevent this decreasing Hamiltonian path; in fact, it makes the ordering almost surely strict.

Consequently, any positive polynomial smoothed theorem for exact PNE computation in unrestricted congestion games must use an additional restriction or a more disciplined pivoting rule. Examples include explicit strategy representations, bounded-restraint conditions, approximation, or shortest-path/best-response dynamics. The construction here marks the boundary for the broadest arbitrary-pivot interpretation.

References

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